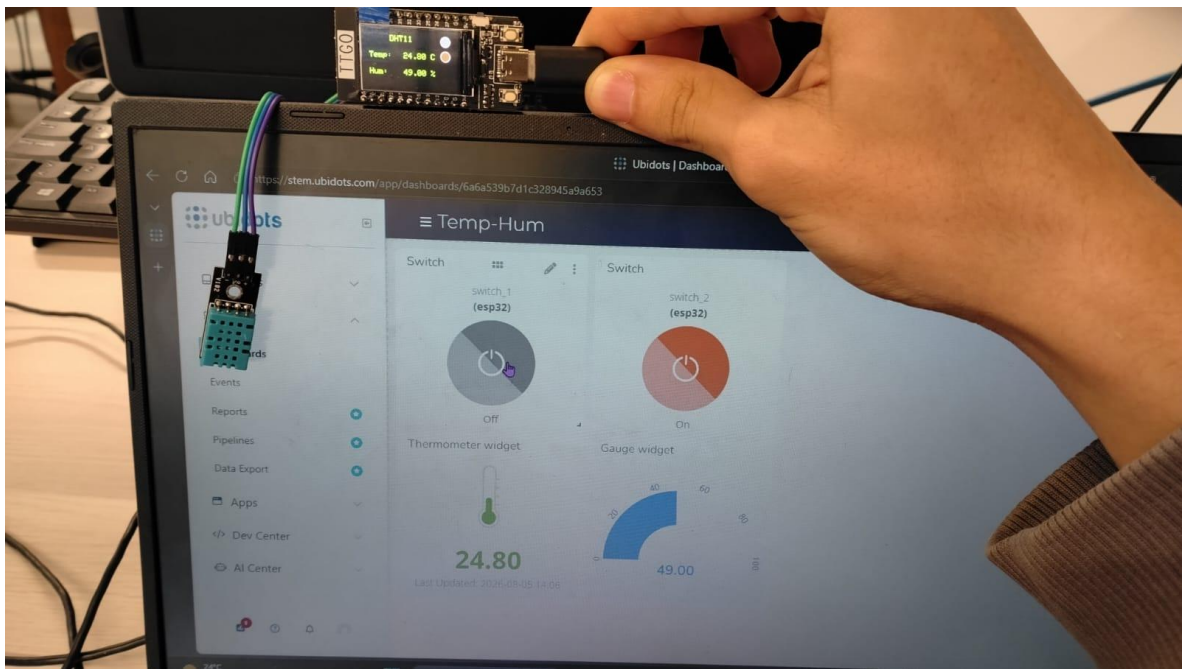
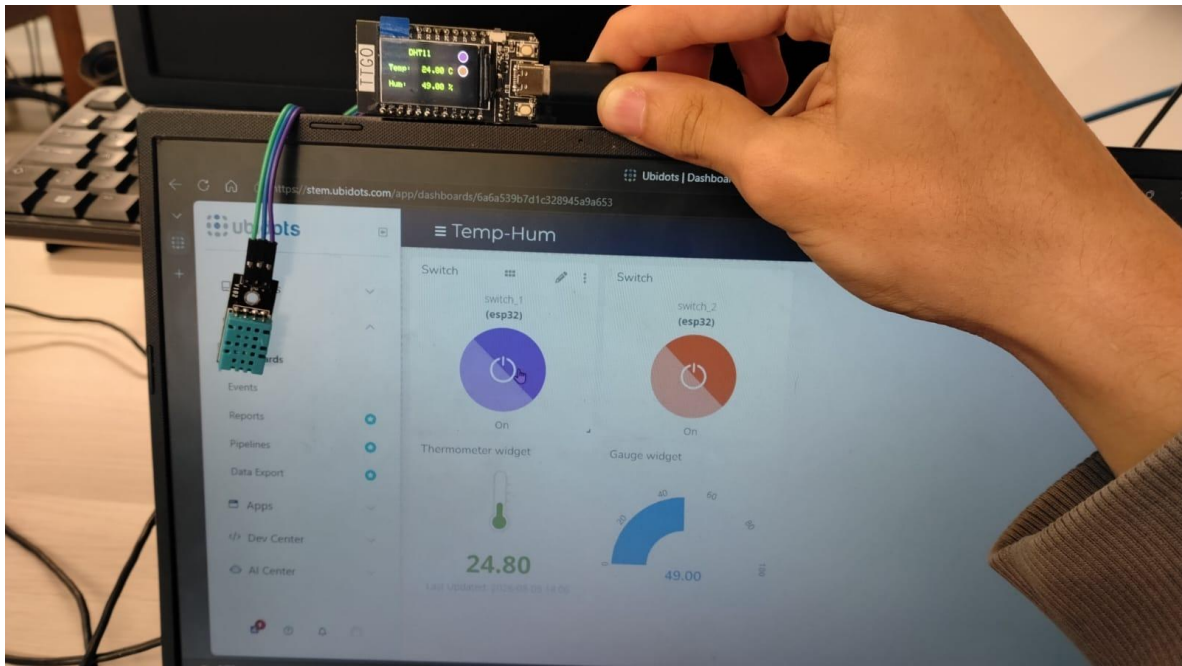
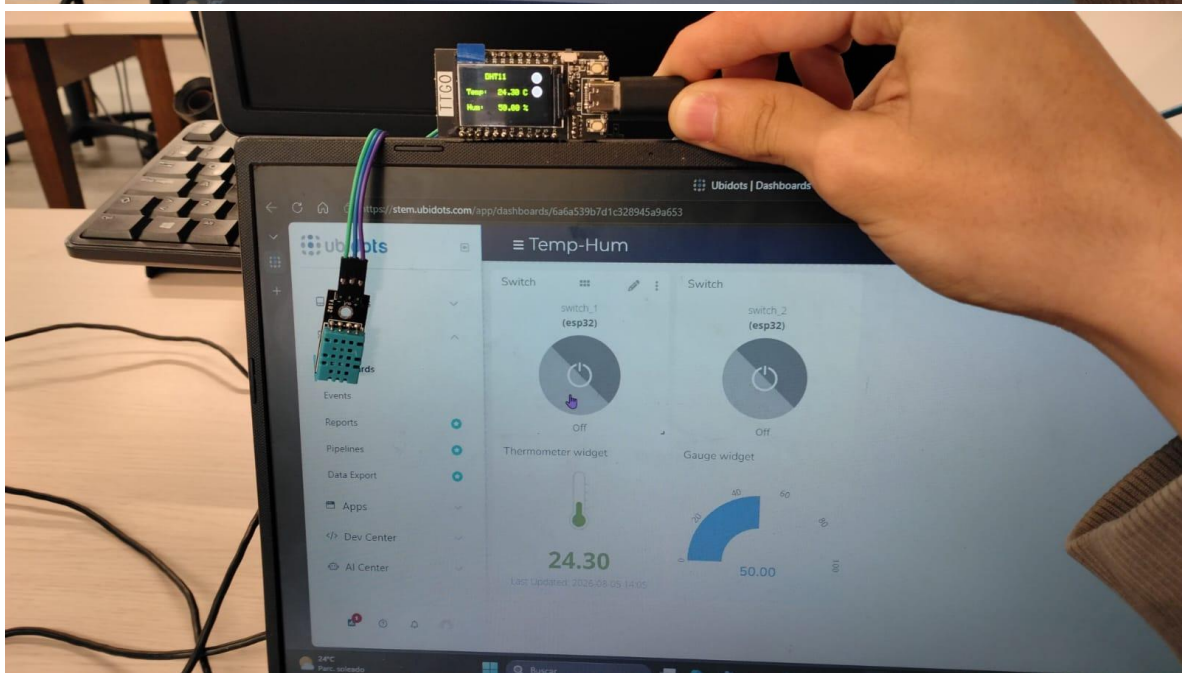
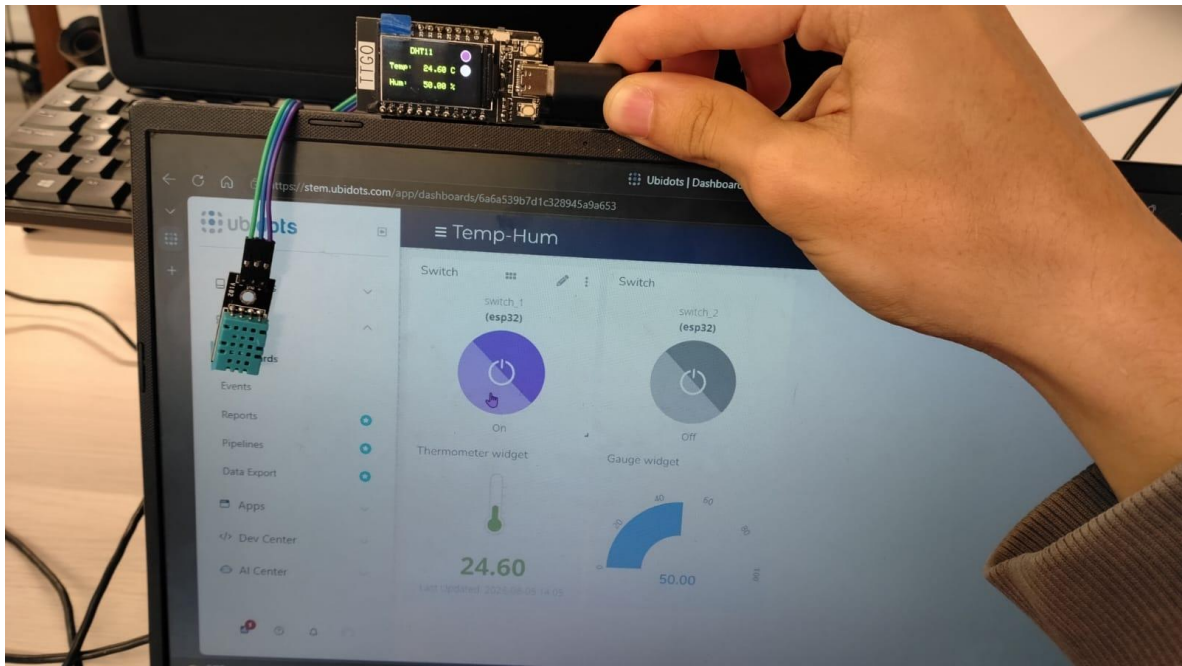






```

/**
 * Libraries
 */
#include "UbidotsEsp32Mqtt.h"
#include <DHT.h>
#include <TFT_eSPI.h>
// =====

/**
 * Constants
 */
const char *UBIDOTS_TOKEN = ██████████
████████████████████████████████████████
const char *WIFI_SSID = "UPBWiFi";
const char *WIFI_PASS = "";
const char *DEVICE_LABEL = "esp32";
const char *VARIABLE_LABEL = "Temperatura";
const char *VARIABLE_LABEL2 = "Humedad";
const char *SUBSCRIBE_DEVICE_LABEL = "esp32";
const char *SWITCH1_VARIABLE_LABEL = "switch_1";
const char *SWITCH2_VARIABLE_LABEL = "switch_2";

volatile bool switch1State = false;
volatile bool switch2State = false;
// =====

const int PUBLISH_FREQUENCY = 5000;
unsigned long timer;
uint8_t analogPin = 33;

#define DHTPIN 27
#define DHTTYPE DHT11
DHT dht(DHTPIN, DHTTYPE);
TFT_eSPI tft = TFT_eSPI();
// =====

Ubidots ubidots(UBIDOTS_TOKEN);

/**
 * Auxiliar Functions
 */

void drawSwitchIndicators() {
    int cx1 = 210, cy1 = 20, r = 12;

```

```

    tft.fillCircle(cx1, cy1, r, switch1State ? TFT_PURPLE :
TFT_DARKGREY);
    tft.drawCircle(cx1, cy1, r, TFT_WHITE);

    int cx2 = 210, cy2 = 55;
    tft.fillCircle(cx2, cy2, r, switch2State ? TFT_RED :
TFT_DARKGREY);
    tft.drawCircle(cx2, cy2, r, TFT_WHITE);
}
// =====

void callback(char *topic, byte *payload, unsigned int length) {
    Serial.print("Message arrived [");
    Serial.print(topic);
    Serial.print("] ");

    char payloadStr[length + 1];
    for (unsigned int i = 0; i < length; i++) {
        payloadStr[i] = (char)payload[i];
        Serial.print((char)payload[i]);
    }
    payloadStr[length] = '\0';
    Serial.println();

    float value = atof(payloadStr);

    String topicStr = String(topic);

    if (topicStr.indexOf(SWITCH1_VARIABLE_LABEL) != -1) {
        switch1State = (value == 1);
        drawSwitchIndicators();
    } else if (topicStr.indexOf(SWITCH2_VARIABLE_LABEL) != -1) {
        switch2State = (value == 1);
        drawSwitchIndicators();
    }
    //
    =====
    =====
}

/**
 * Main Functions
 **/
void setup() {
    Serial.begin(115200);

```

```

dht.begin();
tft.init();
tft.setRotation(1);
tft.fillScreen(TFT_BLACK);
tft.setTextColor(TFT_WHITE, TFT_BLACK);
tft.setTextSize(2);
tft.drawString("Iniciando...", 10, 20);
// =====

ubidots.connectToWifi(WIFI_SSID, WIFI_PASS);
ubidots.setCallback(callback);
ubidots.setup();
ubidots.reconnect();

ubidots.subscribeLastValue(SUBSCRIBE_DEVICE_LABEL,
SWITCH1_VARIABLE_LABEL);
ubidots.subscribeLastValue(SUBSCRIBE_DEVICE_LABEL,
SWITCH2_VARIABLE_LABEL);
// =====

timer = millis();
}

void loop() {
  if (!ubidots.connected()) {
    ubidots.reconnect();
    ubidots.subscribeLastValue(SUBSCRIBE_DEVICE_LABEL,
SWITCH1_VARIABLE_LABEL);
    ubidots.subscribeLastValue(SUBSCRIBE_DEVICE_LABEL,
SWITCH2_VARIABLE_LABEL);
    //
=====
  }

  if ((millis() - timer) > PUBLISH_FREQUENCY) {
    float value = analogRead(analogPin);
    float temperatura = dht.readTemperature();
    float humedad = dht.readHumidity();

    if (!isnan(temperatura) && !isnan(humedad)) {
      ubidots.add(VARIABLE_LABEL, temperatura);
      ubidots.add(VARIABLE_LABEL2, humedad);

      tft.fillScreen(TFT_BLACK);
      tft.setTextColor(TFT_GREEN, TFT_BLACK);

```

```
tft.setTextSize(2);
tft.drawString("DHT11", 65, 10);

tft.setCursor(10, 50);
tft.print("Temp:");
tft.setCursor(100, 50);
tft.print(temperatura);
tft.print(" C");

tft.setCursor(10, 90);
tft.print("Hum:");
tft.setCursor(100, 90);
tft.print(humedad);
tft.print(" %");

drawSwitchIndicators();
}
// =====

ubidots.add("ADC", value);

ubidots.publish(DEVICE_LABEL);
timer = millis();
}

ubidots.loop();
}
```